

LEVEL 4: Linear Search

1. Objective

Search a book by Book ID by checking records sequentially from the first element.

2. Data Structure / Algorithm Used

Array of Structures + Linear Search

3. Problem Statement

Extend the Library Management System by implementing the linear search module. Complete this module independently before integrating it with previous levels.

4. Algorithm / Working

Start at index 0 → compare the key with each Book ID → return the index if matched → return -1 otherwise.

5. C Program

```
#include <stdio.h>
typedef struct { int id; char title[50]; } Book;

int search(Book b[], int n, int key) {
    int i;
    for(i=0; i<n; i++)
        if(b[i].id==key) return i;
    return -1;
}

int main() {
    Book b[]={101,"C Programming"},{105,"Data Structures"},
             {110,"Operating System"};
    int n=3, key, pos;
    printf("Enter Book ID: "); scanf("%d", &key);
    pos=search(b, n, key);
    if(pos!=-1) printf("Found: %d - %s\n", b[pos].id, b[pos].title);
    else printf("Book Not Found\n");
    return 0;
}
```

6. Sample Input

Enter Book ID: 105

7. Sample Output

Found: 105 - Data Structures

8. Complexity

Best: $O(1)$. Average/Worst: $O(n)$. Extra space: $O(1)$.

9. Practice Exercises

- Search by title.
- Count comparisons.
- Handle duplicate Book IDs.

10. Related Viva Questions

- What is linear search?
- Does it require sorted data?
- What is the best case?
- What is the worst case?